Deep Learning Assignment – 1

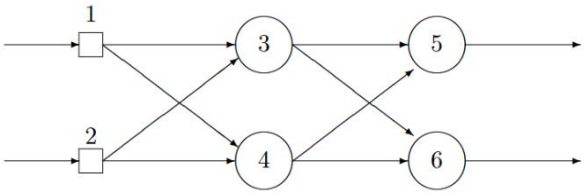
1. Explain various types of activation fuctions with their characteristics , fuctions, graphs , formulas and range.

2. Explain the architecture of Perceptron with neat diagram explaining its componeents.

3. Sketch the architecture of Multi-layer Feed Forward network with Back propagation algorithm and explain its components with updating of Weights and Bias with neat diagrams.

4. Expalin various regularization techniques used in DL?

5.The following diagram represents a Feed-Forward Neural Network with one hidden layer, where a weight on connection between nodes i and j is denoted by wij, such as w13 is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:



$w_{13} = -2$	$w_{35} = 1$
$w_{23} = 3$	$w_{45} = -1$
$w_{14} = 4$	$w_{36} = -1$
$w_{24} = -1$	$w_{46} = 1$

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

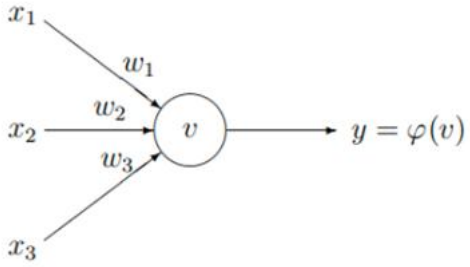
$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

X1	X2
0	0
1	0
0	1

6.Find the updated weights after 1 epoch for the following Neural Network. Use back propagation to update the weights.
Input: [x1, x2] = [2, 3], Weights: [W1, W2] = [0.5, 0.6], Bias: b = 1 and its weight is 0.1, Expected Output: y = 0.5, Activation function: Sigmoid, Loss Function: Mean Squared Error, Learning Rate: η = 0.01.

7. Consider a neural network with 2 input neurons, 2 neurons in the hidden layer, and 1 output neuron. The inputs provided are (1, 0), and the desired output is 0.5. The initial weights for the input to the hidden layer are [0.4, -0.2] and [0.1, 0.3], and for the hidden to output layer are [-0.6, 0.8]. Using the Sigmoid activation function, perform one forward pass and compute the output of the network. Use MSE as error or cost function to update the weights for one epoch.

8. Below is a diagram of a single artificial neuron (unit):



The node has three inputs x = (0, 1, 1) that receive only binary signals (either 0 or 1). Suppose that the weights corresponding to the three inputs have the following values: w1 = 2, w2 = −4, w3 = 1 and the activation of the unit is given by the step-function that produces 1 when v is greater than or equal to 0. If the Input layer had four inputs with w4 = −2, proceed with suitable formula that computes the number of binary input patterns produces '1' for a given number of inputs?

Submit assignment at 618 -A CB and also upload in VTOP.

Last Date to Submit: 29-08-2026 (Saturday)